Right angles on a line

Kind: Problem Date: 12-11-2026

1. The problem

For some points of the form $(k, 7)$, the points $A(2, 3)$, $B(12, 13)$ and $(k, 7)$ form a right-angled triangle. Suppose that C is such a point so that $\triangle ABC$ is right-angled at C. Suppose that D is also such a point so that $\triangle ABD$ is right-angled at D $(D \neq C)$. What is the area of $ACBD$?

What it asks:

1. Points of the form $(k, 7)$ all lie on the horizontal line $y = 7$.
2. "Right-angled at C" means sides CA and CB are perpendicular. Find the two points C and D on $y = 7$ where this happens.
3. Join $A \to C \to B \to D \to A$ and find the area of this quadrilateral.

2. The idea in one line

Perpendicular lines have slopes whose product is -1; then the diagonal DC lies on $y = 7$ and splits $ACBD$ into two triangles with the same base.

3. Key parts

1. **Test one point.** $P(14, 7)$: slope $AP = \frac{4}{12} = \frac{1}{3}$, slope $PB = \frac{6}{-2} = -3$. Product $= -1$, so the angle at P is $90°$. (Watch the sign: from P to B, x goes from 14 down to 12.)
2. **Find every such point.** Replace 14 by k and solve.
3. **Split the area.** DC is horizontal, so each triangle's height is a vertical distance from the line $y = 7$.

4. Worked solution

Finding C and D. For the point $(k, 7)$:

$$\text{slope to } A = \frac{7-3}{k-2} = \frac{4}{k-2}, \qquad \text{slope to } B = \frac{7-13}{k-12} = \frac{-6}{k-12}.$$

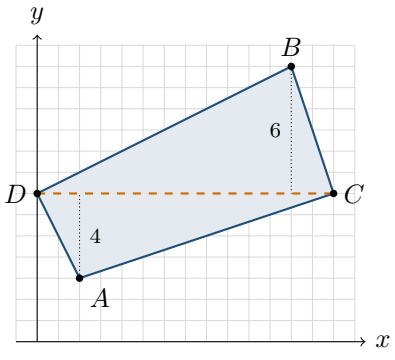
Perpendicular:

$$\frac{4}{k-2} \cdot \frac{-6}{k-12} = -1 \;\Rightarrow\; (k-2)(k-12) = 24 \;\Rightarrow\; k^2 - 14k = 0 \;\Rightarrow\; k(k-14) = 0 \;\Rightarrow\; k = 0 \text{ or } 14.$$

So $C = (14, 7)$ and $D = (0, 7)$.

Area. Base $DC = 14$. Height of $\triangle DCB$: $13 - 7 = 6$. Height of $\triangle DCA$: $7 - 3 = 4$.

$$\text{Area} = \tfrac{1}{2} \cdot 14 \cdot 6 + \tfrac{1}{2} \cdot 14 \cdot 4 = 42 + 28 = \boxed{70}.$$



5. Questions colleagues may ask

1. **Q. Is there a method without slopes?** Yes. An angle subtending a diameter is 90° (Thales' theorem), so C and D lie on the circle with diameter AB: centre $(7, 8)$, radius$^2 = 5^2 + 5^2 = 50$. On $y = 7$: $(k - 7)^2 + 1 = 50$, so $k - 7 = \pm 7$, giving $k = 0$ or 14.
2. **Q. Which other points $(k, 7)$ give a right-angled triangle?** Right angle at A: $k = -2$. Right angle at B: $k = 18$. The question only uses the two with the right angle at the point on $y = 7$.
3. **Q. Can we check the area another way?** The shoelace formula on $A(2, 3), C(14, 7), B(12, 13), D(0, 7)$ gives $\frac{1}{2}\,|280 - 140| = 70$.